



DRY WALL PARTITION



Description:

Dry wall is an interior wall or partition finished in a dry material, usually in the form of prefabricated sheets or panels fastened to galvanized studs and tracks.

Reference Standards:

ASTM A 653/A 653 M-Specification for Steel Sheet, Zinc-Coated (Galvanized or Zinc- Iron Alloy Coated (Galvannealed) by the Hot-Dip Process.

ASTM C 645-09A - Standard Specification for Non-structural Steel Framing Members.

ASTM C 754-99A - Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products.

BS EN 14195 - Metal Framing Components for Gypsum Plasterboard Systems.

Definition of Terms:

Framing Member - metal studs, runners (track) designed to receive screw attached gypsum panel products.

Runner a member designed to receive the ends of metal stud, attached directly to the structural members of the building.

Stud-vertical member in the frame work to which panel boards are fastened.

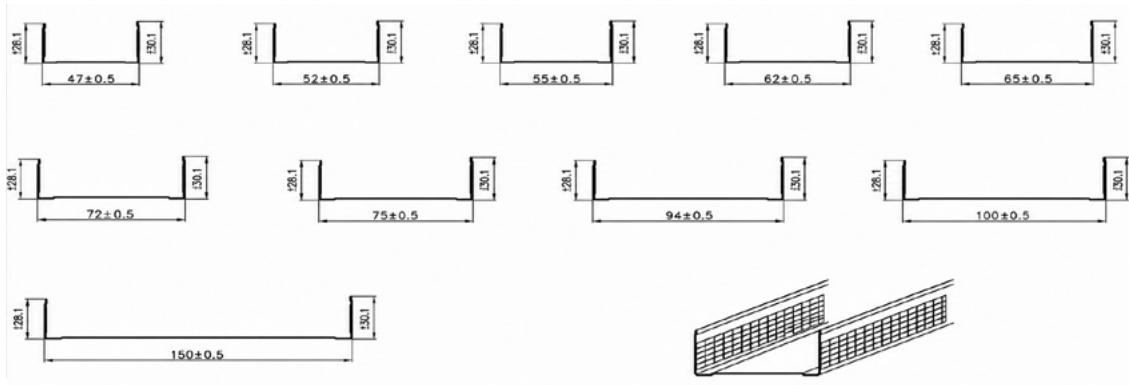
DELIVERY, HANDLING AND STORAGE:

- **Delivery:** Materials are to be delivered in original packages, containers or bundles containing the manufacturers or supplier's identification.
- **Transportation:** In transportation or lifting, handle products with care avoiding excessive stress and preventing damages.
- **Storage:** Store materials under a shaded area and keep them dry and protected against damage due to weather condition, humidity level, job site location, storage condition and surface contamination.



Material:

A. Runner-pre-galvanized G60 and G90; thickness 0.5mm to 0.9mm

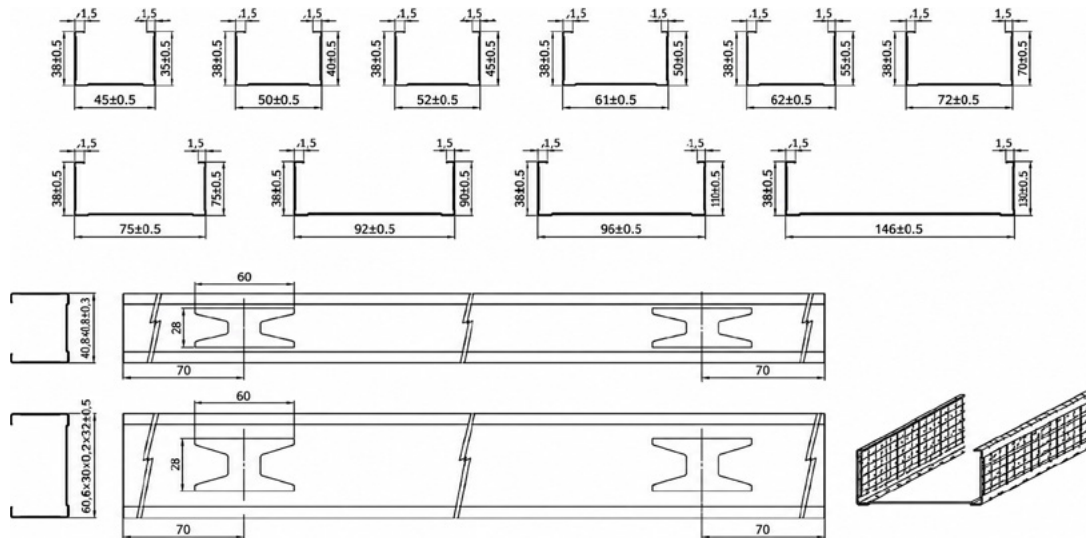


Type Code	Available Thickness in mm	Standard Length in mm	Length (m)	Packing pcs/Blt
T 42	0.45 - 1.00	50.0	3.0	500
T 47	0.45 - 1.00	50.0	3.0	500
T 52	0.45 - 1.00	50.0	3.0	500
T 55	0.45 - 1.00	50.0	3.0	500
T 62	0.45 - 1.00	50.0	3.0	500
T 65	0.45 - 1.00	50.0	3.0	500
T 72	0.45 - 1.00	50.0	3.0	500
T 75	0.45 - 1.00	50.0	3.0	500
T 94	0.45 - 1.00	50.0	3.0	450
T 100	0.45 - 1.00	50.0	3.0	450
T 150	0.45 - 1.00	50.0	3.0	450

*Other lengths can be manufactured as per design.

*Material: Hot Dipped Galvanized steel as per ASTM 653-Zinc Coating Z180-2750

B. Stud-pre-galvanized G60 and G90; thickness 0.5 to 0.9mm



Type Code	Available Thickness in mm	Standard Length in mm	Length (m)	Packing pcs/Blt
S 40	0.45 - 1.00	50.0	3.0	500
S 45	0.45 - 1.00	50.0	3.0	500
S 50	0.45 - 1.00	50.0	3.0	500
S 53	0.45 - 1.00	50.0	3.0	500
S 60	0.45 - 1.00	50.0	3.0	500
S 63	0.45 - 1.00	50.0	3.0	500
S 70	0.45 - 1.00	50.0	3.0	500
S 73	0.45 - 1.00	50.0	3.0	500
S 92	0.45 - 1.00	50.0	3.0	450
S 98	0.45 - 1.00	50.0	3.0	450
S 148	0.45 - 1.00	50.0	3.0	450

*Other lengths can be manufactured as per design.

*Material: Hot Dipped Galvanized steel as per ASTM 653-Zinc Coating Z180-2750



C. Corner Bead -thickness 0.45mm

	Type Available Thickness Length (m)		
	DW Angle	0.35 mm - 45 mm	3.0

D. Edging Bead -thickness 0.45mm

	Type Available Thickness Length (m)		
	L-Type or J Type	0.35 mm - 45 mm	3.0

E. Accessories

Accessories		Dimensions
Gypsum Board	Thickness = 9.5mm, 12.5mm, 15mm, 18mm	
Drywall Screw	Length = 25 mm, 35 mm, 45 mm	
Framing Screw	Length = 11 mm	
Drywall Joint Tape	50 mm	

Installation

1- Preparation:

1. Installation site should be kept clean and dry.
2. It is recommended that a sketch of the layout should be prepared marking all the necessary dimensions and taking into consideration the doors and windows. Partition layouts should be marked accurately at the site using plumb bob, marker and chalk line
3. Calculating Stud and Track Requirements
 - A. Calculate Studs: Based on spacing requirements (ASTM C754-99A), divide the wall length by 600mm (where standard = max. 610mm) in order to calculate the number of studs needed. Add studs for corners and openings.
 - B. Calculating Track: Take the total linear measurement of the wall in meters and multiply by 2 to get the total requirement for floor and ceiling. Track is sold in 3 meter lengths (other lengths can be supplied upon request and consultation with our production department. Add track for door and window headers and sill.

2- Installation of framing system:

Installation shall meet all requirements stated on the project specification and other local statutory requirements.

- A. Cut studs and track to required lengths.
- B. Position tracks starting at the bottom, then to the wall going to the top. Attach tracks using appropriate fastener for the type of material you are attaching the track to (e.g. concrete, steel or timber)
- C. Insert stud at an angle and twist to lock as shown in the figure below. The friction fit will hold the stud in place. Space studs at 600mm (where standard = max. 610mm). Make sure that all the studs are facing in the same direction. For corners, junctions and wall ends, it is recommended to use extra studs and “boxing” the studs as shown in Figure 2. Please note that there are several opinions raised in securing the studs to the tracks, although it is not stated in ASTM, it is in the designers and contractors discretion.

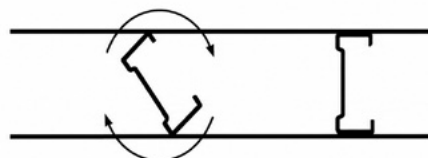


Figure 1 – Installation of stud

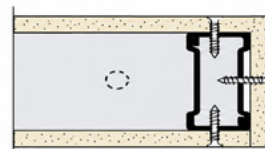


Figure 2 – Wall End (Box Stud)



D. Deflection Track System

It is a part of a wall construction assembly where there is expected vertical movement or deflection and is used to accommodate this movement without damage to the wall finish and system assembly. The two common methods are:

- a. Slotted track - is simpler and faster. This consists of 1" or more as shown in Figure 3. The track is installed as usual but the vertical studs are cut $\frac{1}{2}$ in to 1" shorter and temporarily fastened in place. The fastener will then be removed when the drywall is installed. However, in order not to damage the wall the drywall must not be attached to the vertical legs



Figure 3 - Slotted Track

- b. Double Track (Overtrack) System - The first track is oversized (compared to the stud width) by $\frac{1}{8}$ " and has 2" more vertical legs which are secured to the structure. The second track is a normal track that fits inside the first. Vertical studs will have to be cut short and temporarily secured to the regular track. The temporary fastener will be removed when the drywall is fitted. In this system, the drywall is held down below the oversized track

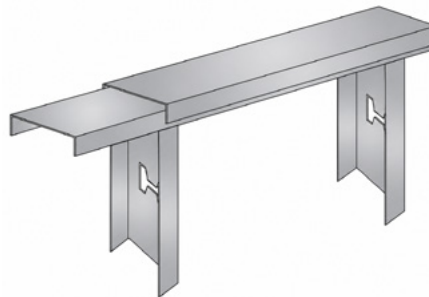


Figure 4 - Double Track (Overtrack) System

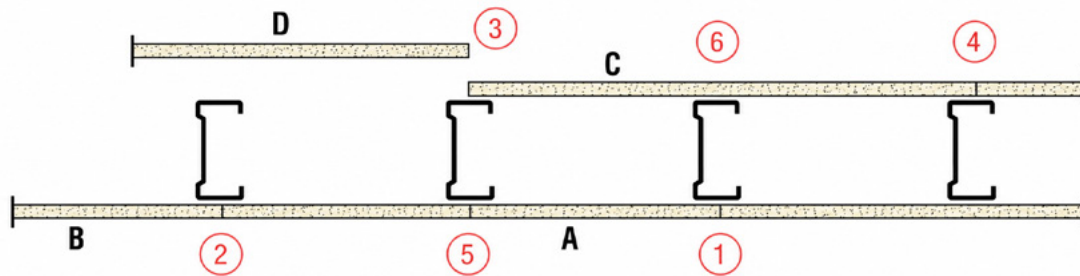
- E. Headers and sill - simply fabricated by cutting a piece of track 6" to 12" longer than the opening. The track is centered in the opening with the excess cut at the vertical legs, and bent 90°, this U-channel metal profile will then be fastened at the correct head or sill height. Note that there also pre-manufactured headers and sills especially used for large projects like hotels

2-Installation of Gypsum boards:

As the face of steel stud can deflect initially during fixing, thus there should be a correct fixing technique to be followed.

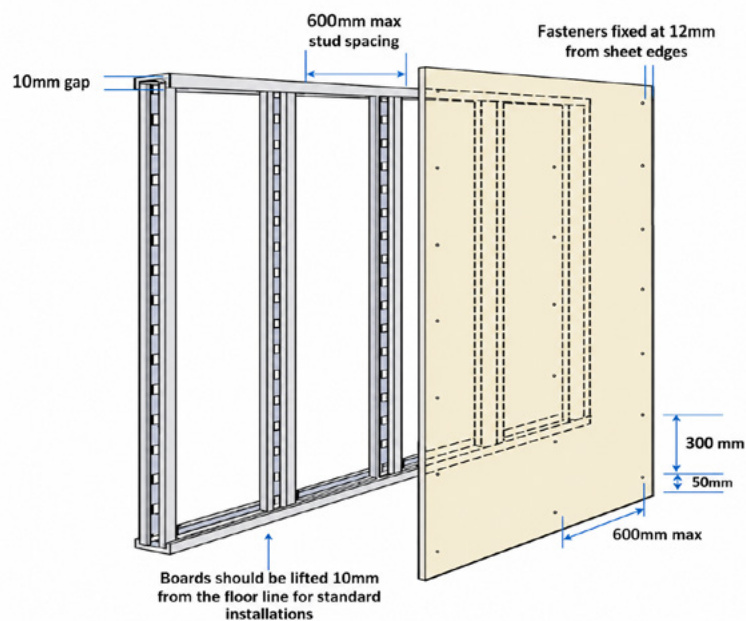
The first board shall be attached to the open side of the stud. Support the stud to avoid twisting and fully screw the board to this stud before continuing to the next. The next sheet can now be screwed to the closed side of the stud. Deflection will be minimal as the first sheet will keep it rigid thus achieving a flat joint.

Board Fixing Sequence



The boards shall be fixed in the sequence shown ABCD. When installing the first side (A&B), fasten the board at the edges only (1&2). The center must not be fastened until the second side (C&D have been installed. Locate board joints on each side of the wall on alternate studs. Fasten the boards in sequence 1-6. This will minimize misalignment and give a higher quality finish.

Installation Details



Installation Details-Continued

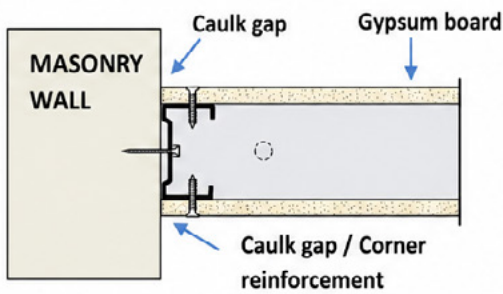


Figure 7 - INTERSECTION – Masonry

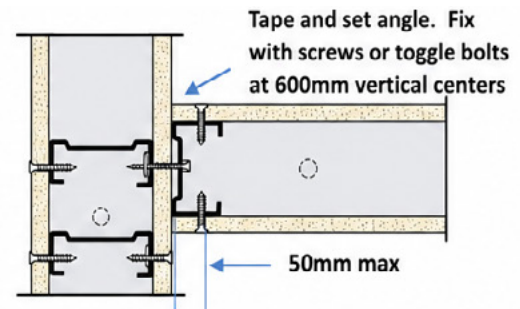


Figure 8 – INTERSECTION – Single Layer Application

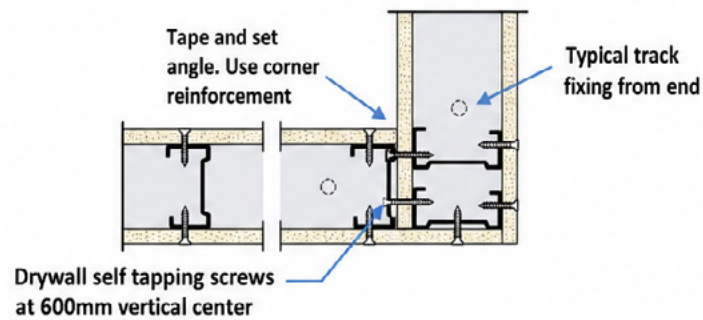


Figure 9 - CORNER - Single Layer Application

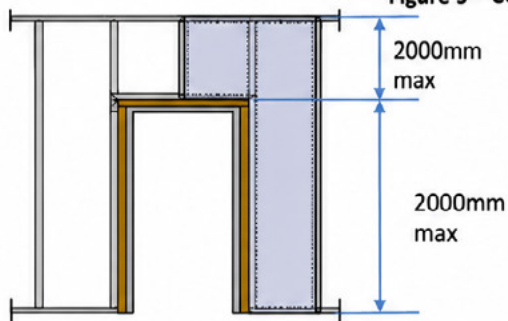


Figure 10 – Single Door

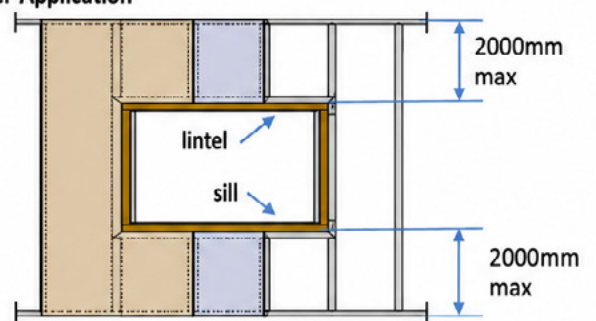


Figure 11 - Windows